

Lognormal Integral Formulation

Problem

Given $p(h)$ as the log-normal distribution PDF and $\alpha(h) = \eta h^k$, compute

$$I(s) = \int_0^s \alpha(h) p(h) dh + \alpha(s) \int_s^\infty p(h) dh$$

where

$$p(h) = \frac{1}{h\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln h - \mu)^2}{2\sigma^2}\right), \quad h > 0$$

and η, μ, σ, k are constants.

Step 1: Substitute $\alpha(h)$ and $p(h)$

$$I(s) = \eta \int_0^s h^k p(h) dh + \eta s^k \int_s^\infty p(h) dh$$

Step 2: Simplify and Change Variable $t = \ln h$

First term:

$$\int_0^s h^k p(h) dh = \int_0^s h^k \frac{1}{h\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln h - \mu)^2}{2\sigma^2}\right) dh = \frac{1}{\sigma\sqrt{2\pi}} \int_0^s h^{k-1} \exp\left(-\frac{(\ln h - \mu)^2}{2\sigma^2}\right) dh$$

Let $t = \ln h$, $dh = e^t dt$, $h = e^t$:

$$\int_0^s h^{k-1} \exp\left(-\frac{(\ln h - \mu)^2}{2\sigma^2}\right) dh = \int_{-\infty}^{\ln s} e^{kt} \exp\left(-\frac{(t - \mu)^2}{2\sigma^2}\right) dt$$

Second term:

$$\int_s^\infty p(h) dh = \int_s^\infty \frac{1}{h\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln h - \mu)^2}{2\sigma^2}\right) dh = \frac{1}{\sigma\sqrt{2\pi}} \int_{\ln s}^\infty \exp\left(-\frac{(t - \mu)^2}{2\sigma^2}\right) dt$$

Step 3: Final Analytical Solution

Combine and complete the square in the exponent for the first term:

$$I(s) = \frac{\eta}{\sigma \sqrt{2\pi}} \left[\int_{-\infty}^{\ln s} \exp\left(k t - \frac{(t - \mu)^2}{2\sigma^2}\right) dt + s^k \int_{\ln s}^{\infty} \exp\left(-\frac{(t - \mu)^2}{2\sigma^2}\right) dt \right]$$

The first integral can be rewritten as:

$$\int_{-\infty}^{\ln s} \exp\left(k t - \frac{(t - \mu)^2}{2\sigma^2}\right) dt = e^{\left\{k\mu + \frac{1}{2}k^2\sigma^2\right\}} \int_{-\infty}^{\ln s} \exp\left(-\frac{(t - \mu - k\sigma^2)^2}{2\sigma^2}\right) dt$$

So,

$$I(s) = \frac{\eta}{\sigma \sqrt{2\pi}} \left[e^{\left\{k\mu + \frac{1}{2}k^2\sigma^2\right\}} \int_{-\infty}^{\ln s} \exp\left(-\frac{(t - \mu - k\sigma^2)^2}{2\sigma^2}\right) dt + s^k \int_{\ln s}^{\infty} \exp\left(-\frac{(t - \mu)^2}{2\sigma^2}\right) dt \right]$$

These integrals are proportional to the CDF and complementary CDF of the normal distribution.

Python code

See `Lognormal Integral Eval.py` for a Python implementation.